

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 2_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

John is a fitness trainer, and he wants to use the BMI calculator to assess the body mass index of his clients. He has a list of clients based on their height and weight.

John plans to write a program to quickly determine the BMI and provide a classification for each client.

If BMI is less than 18.5, the program will classify it as "Underweight" If BMI is between 18.6 and 24.9, the program will classify it as "Normal Weight" If BMI is between 25.0 and 29.9, the program will classify it as "Overweight" If BMI is 30.0 or higher, the program will classify it as "Obese"

Note: Formula to calculate BMI = $\text{weight}/(\text{height}*\text{height})$

Input Format

The first line of input consists of a double value, representing the height of the person in meters.

The second line consists of a double value, representing the weight of the person in kilograms.

Output Format

The first line of output prints "BMI: " followed by a double (rounded to two decimal places) representing the calculated BMI.

The second line prints "Classification: " followed by a string indicating the BMI category (Underweight, Normal Weight, Overweight, or Obese).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1.2

45.2

Output: BMI: 31.39

Classification: Obese

Answer

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);
```

```
        // Read height and weight (two doubles)  
        double height = scanner.nextDouble();  
        double weight = scanner.nextDouble();
```

```
        // Calculate BMI  
        double bmi = weight / (height * height);
```

```
        // Round BMI to 2 decimal places  
        // Using String.format or printf for formatting output  
        System.out.print("BMI: " + String.format("%.2f", bmi) + " ");
```

```
// Classification based on BMI ranges
if (bmi < 18.5) {
    System.out.println("Classification: Underweight");
} else if (bmi >= 18.6 && bmi <= 24.9) {
    System.out.println("Classification: Normal Weight");
} else if (bmi >= 25.0 && bmi <= 29.9) {
    System.out.println("Classification: Overweight");
} else if (bmi >= 30.0) {
    System.out.println("Classification: Obese");
}

scanner.close();
}
```

Status : Correct

Marks : 10/10